

Future Research Statement

Towards Adaptive Foundation Models through Inference-Time Intelligence

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1 Research Vision

Artificial intelligence is entering a new phase in which progress will depend less on continually increasing model size and more on developing computational mechanisms that enable pretrained foundation models to adapt efficiently after pretraining. Although recent foundation models possess remarkable semantic and multimodal capabilities, their adaptation to downstream tasks remains largely dependent upon parameter optimisation through fine-tuning, prompt learning, or reinforcement learning. As these models continue to increase in scale, optimisation-centric adaptation becomes increasingly expensive, task-specific, and difficult to deploy.

My long-term research vision is to develop adaptive foundation models capable of coordinating, reasoning, and learning at inference time while preserving the knowledge acquired during large-scale pretraining. I believe that inference-time computation will become an increasingly important paradigm for artificial intelligence, enabling models to exploit complementary representations without continual retraining. Through this research programme, I seek to establish computational principles that unify visual perception, multimodal reasoning, and reinforcement learning within a common framework of inference-time intelligence.

2 Research Theme I: Inference-Time Coordination for Foundation Models

My doctoral research establishes the first stage of this programme by introducing inference-time coordination as a computational framework for adapting pretrained representations under limited supervision. Future work will generalise this framework beyond semantic segmentation to a broader range of perception and reasoning problems.

The immediate objective is to investigate how heterogeneous pretrained representations—including visual embeddings, language representations, geometric priors, and memory structures—can be coordinated through principled inference-time operators. Rather than developing increasingly

specialised adaptation modules, I aim to construct general coordination algorithms that remain architecture-agnostic and transferable across multiple foundation models.

This research will investigate theoretical properties of inference-time coordination, including representation consistency, uncertainty propagation, calibration, and efficient evidence aggregation. The resulting framework has the potential to provide a unified perspective on adaptation across computer vision, multimodal learning, and large-scale representation learning.

3 Research Theme II: Reinforcement Learning for Multimodal Reasoning

The second component of my future research focuses on reinforcement learning as a mechanism for improving reasoning rather than representation learning alone. Recent reasoning models have demonstrated that reinforcement learning can substantially enhance mathematical reasoning and language generation. Extending these advances to multimodal systems remains an important open research challenge.

My research will investigate process-level reward design for multimodal foundation models capable of integrating visual perception and symbolic reasoning. Particular emphasis will be placed on reward functions that evaluate intermediate reasoning steps through semantic consistency, visual grounding verification, spatial correspondence, and evidence completeness. Rather than relying exclusively on final-answer correctness, these reward functions will encourage reasoning trajectories that remain faithful to the underlying visual evidence.

This direction naturally extends my doctoral work by replacing static inference-time coordination with adaptive reasoning policies that learn to coordinate multiple sources of pretrained knowledge during sequential decision making.

4 Research Theme III: Adaptive Multimodal Intelligence

The third research theme investigates the integration of perception, reasoning, and interaction within unified multimodal systems. Current vision-language models have demonstrated remarkable capabilities but continue to exhibit limitations in long-horizon reasoning, uncertainty estimation, and spatial understanding. I aim to develop computational frameworks that enable these systems to combine semantic, geometric, and temporal information more effectively.

This research will investigate multimodal agents capable of integrating visual observations, language instructions, structured knowledge, and environmental feedback within a common inference-time framework. Potential application domains include autonomous scientific discovery, intelligent medical image analysis, robotics, remote sensing, and decision-support systems where reliable reasoning under limited supervision is essential.

5 Long-Term Research Programme

The long-term objective of this research is to establish Inference-Time Intelligence as a general computational paradigm for adaptive artificial intelligence. Rather than viewing perception, reasoning, and reinforcement learning as independent research problems, I propose a unified perspective in which pretrained foundation models coordinate complementary representations through efficient inference-time computation.

This programme seeks to address several fundamental questions:

How can pretrained representations be coordinated without parameter optimisation? How can reinforcement learning improve multimodal reasoning while preserving pretrained knowledge? How should heterogeneous semantic, geometric, and language representations interact during inference? Can inference-time coordination provide a computationally efficient alternative to continual retraining?

Addressing these questions has the potential to influence the development of future foundation models that are more adaptive, interpretable, and computationally efficient.

6 Collaboration and Impact

I am particularly interested in collaborating with research groups working at the intersection of computer vision, multimodal learning, reinforcement learning, and foundation models. My background combines theoretical modelling, algorithm development, empirical evaluation, and large-scale software implementation, enabling me to contribute both fundamental research and practical system development.

I view a postdoctoral appointment as an opportunity not only to deepen my expertise in adaptive artificial intelligence but also to contribute to collaborative research programmes addressing some of the most significant open challenges in multimodal reasoning and efficient foundation-model adaptation. Through interdisciplinary collaboration, I hope to develop algorithms that are scientifically rigorous, practically impactful, and broadly applicable across diverse real-world domains.

7 Closing Vision

My long-term ambition is to establish an independent research laboratory dedicated to Inference-Time Intelligence, investigating how foundation models can coordinate representations, reason across modalities, and adapt efficiently without continual retraining. By integrating computer vision, multimodal learning, reinforcement learning, and representation learning within a unified computational framework, I aim to contribute to the development of adaptive AI systems capable of robust perception, trustworthy reasoning, and efficient decision making across complex environments.